Digital Material Passport

ID **DMP-METAL-003** - *Version* **1.0.0**
Issue Date **2025-05-15** - *Certificate Type* **EN 10204 3.1**

Manufacturer	Customer	Business Transaction
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com	**Precision Aerospace Inc.** Aviation Boulevard 789 Mountain View 94043 US materials@precision-aero.example.com	**Order** PO-34567 · **500 kg** Pos. 3 · 2025-04-25 **Delivery** DN-89012 · **500 kg** Pos. 1 · 2025-05-14

Product

Product Name	Aluminum Alloy 7075-T6	*Name (AA)*	7075-T6
Batch ID	H-43210-01	*Number (UNS)*	A97075
Surface Condition	Rolled	*Shape*	Plate - Width 1000 mm - Thickness 10 mm - Length 2000 mm
Weight	500 kg	*Product Norm*	AMS 4045 (2023)
Production Date	2025-05-12		
Country of Origin	DE		

Chemical Analysis

Heat Number **H-43210** - *Melting Process* **VAR** - *Casting Method* **VacuumCasting** - *Casting Date* **2025-05-10** - *Sample Location* **Ladle**

Symbol	*Unit*	*Min*	*Max*	*Actual*
Al	%			89.7
Zn	%	5.1	6.1	5.6
Mg	%	2.1	2.9	2.4

Symbol	*Unit*	*Min*	*Max*	*Actual*
Cu	%	1.2	2	1.5
Cr	%	0.18	0.28	0.22
F1		0.18	0.28	0.22

F1 = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.22

Mechanical Properties

Property	*Symbol*	*Actual*	*Minimum*	*Maximum*	*Method*	*Status*
Tensile Strength[1]	Rm	570 / 572 / 574 MPa - Average 572 - Min 570 - Max 574 - Std Dev 2	530 MPa		ASTM E8	✓
0.2% Yield Strength[1]	Rp0.2	503 / 505 / 507 MPa - Average 505 - Min 503 - Max 507 - Std Dev 2	480 MPa		ASTM E8	✓
Elongation[1]	A	10.8 / 11.2 / 11 % - Average 11 - Min 10.8 - Max 11.2	10 %		ASTM E8	✓

[1] 3 specimens tested

Physical Properties

Property	*Symbol*	*Actual*	*Target/Min*	*Maximum*	*Method*	*Status*
Density	ρ	2.81 g/cm³	2.81 g/cm³		ASTM B311	✓
Coefficient of Thermal Expansion Temperature range: 20-100°C	α	23.4 10^{-6}/K	23.5 10^{-6}/K		ASTM E228	✓
Thermal Conductivity At 25°C	λ	130 W/(m·K)	120 W/(m·K)		ASTM E1461	✓
Specific Heat Capacity At 25°C	cp	862 J/(kg·K)	860 J/(kg·K)		ASTM E1269	✓
Electrical Resistivity At 20°C	ρ_e	0.0538 µΩ·m		0.055 µΩ·m	ASTM B193	✓
Poisson's Ratio	ν	0.33	0.33		ASTM E132	✓
Melting Range	Tm	477 - 635 °C	475 - 635 °C		ASTM E1142	✓
Relative Magnetic Permeability	µr	1.00002		1.0001	ASTM A342	✓
Surface Roughness	Ra	0.8 µm		1.6 µm	ISO 4287	✓
Emissivity At 25°C	ε	0.09		0.11	ASTM E408	✓
Surface Tension	γ	0.875 N/m	0.87 N/m		ASTM D971	✓
Diffusion Coefficient	D	2.3E-9 m²/s	2.2E-9 m²/s		ASTM E1559	✓

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with AMS 4045 and the specified test methods. All results are within the specified limits.

Validated by **Elsa Müller**, Materials Engineer, Quality Assurance - 2025-05-15